

100 Days of Azure DevOps

Day 52 Handout — App images belong in a private registry — Docker Hub is for bases you meant to be public

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — ACR vs Docker Hub (same docker push, different closet)

Dimension	ACR (azurecr.io)	Docker Hub
Audience	Private registry in your subscription. Default deny to the world	Public-by-habit for tags unless the repo is paid-private
Name	Globally unique. myacr.azurecr.io/myapp:tag	docker.io/library/... or your Hub namespace
Auth	az acr login / AcrPush+AcrPull RBAC / WIF. Admin user is a password in a drawer	Hub credentials. Fine for public bases; wrong for app bits + .env accidents
SKU	Basic (lab), Standard, Premium (geo-replication, private link, retention)	Hub plan limits. Not an Azure resource you can RBAC
Tasks	acr build / ACR Tasks can build in Azure. Optional literacy	Hub automated builds are a different product. Not today's lab

Architecture B — decide where the bits live

Constraint	Default	Move when
Base images	Pull pinned public bases (node:20-alpine)	Air-gap / policy forbids public pulls — import into ACR
App image myapp	Push to ACR. Never Hub	There is no valid lab exception
Login	az acr login -n <uniqueacr> (AAD)	Pipeline tomorrow: service connection with AcrPush, not admin user
SKU	Basic for rg-day52	Premium only for private link / geo-repl you can name
Retention	Delete tags you do not need today	Untagged manifests pile up and still bill

Architecture C — pitfalls that leak an app into the thrift store

Pitfall	What you see	Fix
Push to docker.io by habit	docker push myapp:latest goes public-ish	Retag to <acr>.azurecr.io/myapp:day52 first
Admin user enabled for convenience	Two passwords in Portal; they leak in screenshots	admin-enabled false; AcrPush on an identity
latest as the only tag	Cannot say which bits ACI will pull tomorrow	Tag day52 (and later Build.BuildId). latest is optional pointer
Name collision	az acr create fails; registry names are global	Pick another unique name. Do not reuse a registry you do not own
Leaving every tag	Basic ACR still bills; untagged manifests linger	az acr repository delete / untag what the lab does not need

One-liner to remember

ACR is a private closet for images — public Docker Hub is the thrift store. Put app bits in the closet; keep Hub for bases you intended to be public.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure subscription only. Resource group rg-day52. Personal Azure DevOps project azure-100-labs.

- 1 az group create -n rg-day52 -l eastus. az acr create -g rg-day52 -n <uniqueacr> --sku Basic --admin-enabled false. If the name is taken, pick another — do not hijack a registry.
- 2 az acr login -n <uniqueacr>. Confirm docker is talking to that login server, not docker.io.
- 3 docker tag myapp:day51 <uniqueacr>.azurecr.io/myapp:day52 then docker push <uniqueacr>.azurecr.io/myapp:day52. Optional second tag :latest as a pointer only.
- 4 Portal → Container registries → Repositories, or az acr repository show-tags -n <uniqueacr> --repository myapp -o table. Copy the digest, not only the tag.
- 5 Do not enable admin user. Do not docker push the app image to Docker Hub. Delete extra tags you created as experiments.
- 6 Write docs/acr-day52.md: registry name, tag, digest, and the one-line rule (Hub for bases, ACR for myapp). Leave the registry for Day 53 unless cost is tight — then note you will recreate.

Starter CLI — private tag, then push

```
az group create -n rg-day52 -l eastus
az acr create -g rg-day52 -n <uniqueacr> --sku Basic --admin-enabled false
az acr login -n <uniqueacr>
docker tag myapp:day51 <uniqueacr>.azurecr.io/myapp:day52
docker push <uniqueacr>.azurecr.io/myapp:day52
az acr repository show-tags -n <uniqueacr> --repository myapp --detail -o table
```

Done checklist

- Basic ACR exists on the personal subscription; admin user is off
- myapp:day52 is in <acr>.azurecr.io, not on Docker Hub
- Can explain tag vs digest and why latest is not the promotion ID
- docs/acr-day52.md recorded the digest

Tomorrow — Day 53

Build and push the image from a pipeline — humans stop tagging on Fridays